ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

TECHNOLOGY

PAPER 1 · PRACTICE D · MARKING GUIDELINE

MARKS: 75

Marking guideline for the Academic Central Grade 8 Technology paper 1 · practice d.

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words and other correct examples. In calculations give method marks even if the final answer is wrong.

QUESTION 1

1 1.1 A ✓ 1.2 C ✓ 1.3 B ✓ 1.4 D ✓ 1.5 B ✓ 1.6 A ✓ 1.7 B ✓ 1.8 C ✓ 1.9 C ✓ 1.10 A ✓ [10]

QUESTION 2

2.1 (a) Frame ✓ (b) Shell ✓ (c) Solid ✓ (3)

2.2 (a) Tension ✓ (b) Compression ✓ (c) Torsion ✓ (d) Shear ✓ (4)

2.3 (a) A gusset is a (triangular) plate fixed over a joint ✓; it holds the parts firmly together and stops the joint from bending or moving ✓. (4)
(b) Folding a thin sheet into ridges ✓ makes it much stiffer, so it can carry loads without bending ✓.

2.4.1 Any sensible brief ✓✓, e.g. "Design and make a stand that holds a cell phone upright on a desk so that a learner can watch videos without holding the phone." (2)

2.4.2 Any THREE sensible specifications ✓✓✓, e.g. fits phones up to 8 cm wide; holds the phone at an angle of about 60°; made from recycled materials; costs less than R30; does not scratch the phone; leaves space for the charging cable. (3)

2.4.3 Any TWO ✓✓: a wide base; a low centre of gravity (heavy base); non-slip rubber feet; supports at the back (triangle shape) (2)

[18]

QUESTION 3

3.1 (a) Resistance to scratching, denting or wear ✓ (b) The ability to carry a load without breaking ✓ (c) Breaks or shatters easily when hit, without bending ✓ (3)

3.2 Ferrous, any TWO ✓✓: iron; mild steel; stainless steel; cast iron. Non-ferrous, any TWO ✓✓: aluminium; copper; brass; zinc; tin; lead; gold. (4)

3.3 (a) Iron reacts with oxygen and water (moisture) ✓✓ (4)
(b) Any TWO ✓✓: paint it; galvanise it (zinc coating); oil or grease it; powder-coat it; keep it dry

3.4 It is the recycling (resin identification) code that shows which type of plastic it is ✓; it helps people and recycling plants to sort the plastics for recycling ✓. (2)

3.5 The metal pot conducts heat well, so food cooks quickly ✓; plastic is a poor conductor (insulator) of heat, so the handle stays cool and is safe to hold ✓. (2)

[15]

QUESTION 4

4.1.1 $\text{Gear ratio} = \text{driven teeth} \div \text{driver teeth} = 36 \div 12 \checkmark = 3:1 \checkmark$ (2)

4.1.2 $\text{Speed of C} = 90 \times 12 \div 36 \checkmark = 30 \text{ r/min} \checkmark$ (2)

4.1.3 Clockwise ✓ (A clockwise, B anticlockwise, C clockwise) (1)

4.1.4 No ✓ – the speed depends only on the driver and the driven gear; the idler only changes the direction ✓ (2)

4.1.5 To increase force (torque) ✓, because the driven gear turns more slowly than the driver ✓ (2)

4.2 Use ✓, e.g. tuning pegs of a guitar, gate openers, conveyors, winches. Advantage ✓, e.g. a very large speed reduction in a small space; it cannot be turned backwards by the load, so it locks in position. (2)

4.3 Cam (an off-centre or pear-shaped wheel) on a shaft ✓; follower resting on the cam ✓; rotary motion of the cam shown ✓; reciprocating (up-and-down) motion of the follower shown ✓ (4)

4.4 Lubrication reduces friction between moving parts ✓, which reduces wear and heat, so the machine lasts longer and works more efficiently ✓. (2)

[17]

QUESTION 5

5.1.1 Battery ✓; two switches in parallel ✓; buzzer in series with the switches ✓; correct symbols and a closed circuit ✓ (4)

5.1.2 Inputs listed in all four combinations ✓; outputs ✓✓✓ (deduct one for each wrong output): (4)

Front door (A)	Back door (B)	Buzzer
0	0	0
0	1	1
1	0	1
1	1	1

5.2 Any TWO with a use ✓✓, e.g. push-button switch – doorbell; toggle (light) switch – room lights; slide switch – torch; reed switch – burglar alarm on a window; micro switch – fridge door light (2)

5.3 $\text{Energy per day} = 2 \times 0{,}5 = 1 \text{ kWh} \checkmark; \text{for 30 days} = 30 \text{ kWh} \checkmark; \text{cost} = 30 \times \text{R3,50} = \text{R105} \checkmark$ (3)

5.4 Any TWO ✓✓: never touch switches or plugs with wet hands; do not overload plugs; do not use damaged cords; never push objects into sockets; switch off before unplugging; keep appliances away from water (2)

[15]

TOTAL: 75

END OF MARKING GUIDELINE